Term Project Option 4

Hyperspectral Image Inter-band Prediction

1. Introduction

In this project, you are asked to develop an efficient hardware accelerator for an inter-band predictor used in hyperspectral image compression. Hyperspectral images are taken from the satellite to capture the geographical information of the landscape. Each set of hyperspectral images, as shown in Figure 1, consists of 224 bands of image obtained by using different wavelengths of infrared. The pixel value is 16-bit long.

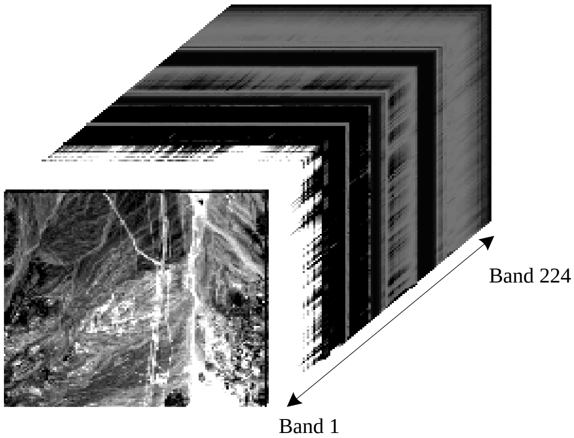

Figure 1. hyperspectral images

2. Linear 1-tap predictor

Due to high data correlation across adjacent bands, an inter-band predictor is devised to eliminate the data redundancy. The target predictor is a linear 1-tap predictor using information from one reference band (band *i*-1 with respect to the current band *i*). The prediction formula is described as

$$\hat{y} = \alpha(x - m_x) + m_y \,. \tag{1}$$

Refer to the context window definitions given in Figure 2, assume pixel y in the current band is to be predicted. Pixel w and x in reference bands have coincident locations with pixel y.

	x_{20}	x_{18}	x_{17}	x_{19}	x_{21}
x_{16}	x_{13}	x_9	x_7	x_6	x_{10}
x_{15}	x_{12}	x_8	x_3	x_2	x_4
x_{14}	x_{11}	x_5	x_1	x	

(b)

	y_{20}	y_{18}	y_{17}	y_{19}	y_{21}
y_{16}	y_{13}	y_9	y_7	y_6	y_{10}
y_{15}	y_{12}	y_8	y_3	y_2	y_4
y_{14}	y_{11}	y_5	y_1	y	

(c)

Fig.2. definition of context windows (a) reference band i-1, (b) current band i

The parameters in Eq. (1) can be calculated as follow

$$\alpha = \frac{\sigma_{xy}}{\sigma_x^2} \tag{2}$$

where

$$\sigma_x^2 = E\left\{x^2\right\} - m_x^2 = M\sum_{i=1}^{M} x_i^2 - \left(\sum_{i=1}^{M} x_i\right)^2 \tag{3}$$

$$\begin{aligned} \sigma_{xy} &= E\left\{xy\right\} - m_x m_y \\ &= M\sum_{i=1}^{M} x_i y_i - \sum_{i=1}^{M} x_i \sum_{i=1}^{M} y_i \end{aligned} \tag{4}$$

$$m_x = \frac{1}{M}\sum_{i=1}^{M} x_i \tag{5}$$

M (= 20) is the size of the context window. Assume the size of each image is 512 by 614 and you have 223 bands (i = 2 ~ 224) of image to be predicted.

Note : m_x and m_y are the means (expected values) of the random variables x and y.

3. Design Guidelines

All 224 bands of images are initially stored in an external memory capable of communicating with on chip memory modules at a data bandwidth no less than 16-bit/clock. (You may specify the required bandwidth subject to your own design but the number should be kept minimal). The data will be accessed in a row-wise manner. The total storage size of the on chip memory modules should not exceed the size of 3 bands of image. (You may use multiple memory modules within the chip to increase the internal data access bandwidth but the limitation on the total memory size should be observed). The calculated prediction results $\hat{y}$ should be written back to the external memory modules. A possible system block diagram of this design is shown in Figure 3.

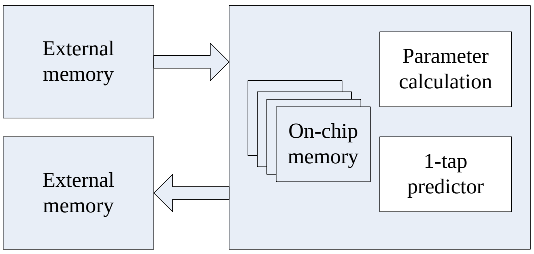

Figure 3. System block diagram

There is NO limit on the amount of hardware employed to accelerate the computations. A minimum computing parallelism requirement is that you should perform predictions on at least two pixels concurrently. Besides the memory access issue, simplification on the computing complexity is the center of your design efforts. Please note that there exists some form of computation overlap in calculating Eq (3)~(4) for two adjacent pixels. Exploring such computing redundancy and minimizing the memory access bandwidth (i.e.

data reuse) are the keys to the success of your design. Please synthesize your design using TSMC 0.18um process technology.

4. Design Merits

The effectiveness of your design will be evaluated based on the following performance indexes:

- Total logic gate count – for the data path and the control path
- Total on-chip memory size – please provide your memory configurations and the equivalent gate count as well
- Degree of computing parallelism – the number of concurrent pixel predictions
- Internal memory access bandwidth – in the unit of MB/sec
- External memory access bandwidth – in the unit of MB/sec
- Maximum working frequency – in the unit of MHz
- Clock cycles needed to complete the prediction of one image
- Throughput rate – number of images can be processed per second
- Power consumption at maximum working frequency
- Energy needed to complete the prediction of one image – = power/throughput rate

5. Hand in requirements

- A report in MS word format highlights your architecture design, data path design, memory configuration, simulation and synthesis results, and the performance indexes stated above. You are also suggested to describe the algorithm mapping techniques applied in this project.
- A presentation in power point format to demonstrate the merits of your design.
- A tar file containing all design files and documents for the uploading purpose.

6. Supplementary information

The hyperspectral image data as well as a Matlab code for the prediction will be provided on the course web site (available later).